

Reproducibilidad con Impacto

Lars Vilhuber

2026-01-23



Follow along



larsvilhuber.github.io/presentation-2026-01/ (PDF)

Reproducibilidad con Impacto

Cómo la Ciencia Abierta Puede Cambiar el Mundo



Lars Vilhuber
Cornell University



What is this?

0005b60	362c	3238	2e32	3433	320a	3230	2d35	3031
0005b70	332d	2c31	3836	3034	322e	0a30	3032	3532
0005b80	312d	2d31	3330	362c	3538	2e31	3739	320a
0005b90	3230	2d35	3131	302d	2c34	3736	3137	352e
0005ba0	0a35	3032	3532	312d	2d31	3530	362c	3937
0005bb0	2e36	3932	320a	3230	2d35	3131	302d	2c36
0005bc0	3736	3032	332e	0a32	3032	3532	312d	2d31
0005bd0	3730	362c	3237	2e38	3038	320a	3230	2d35
0005be0	3131	312d	2c30	3836	3233	342e	0a33	3032
0005bf0	3532	312d	2d31	3131	362c	3438	2e36	3136
0005c00	320a	3230	2d35	3131	312d	2c32	3836	3035
0005c10	392e	0a32	3032	3532	312d	2d31	3331	362c
0005c20	3337	2e37	3934	320a	3230	2d35	3131	312d
0005c30	2c34	3736	3433	312e	0a31	3032	3532	312d
0005c40	2d31	3731	362c	3736	2e32	3134	320a	3230
0005c50	2d35	3131	312d	2c38	3636	3731	332e	0a32
0005c60	3032	3532	312d	2d31	3931	362c	3436	2e32
0005c70	3631	320a	3230	2d35	3131	322d	2c30	3536
0005c80	3833	372e	0a36	3032	3532	312d	2d31	3132



If you I tell you that...



If I give you a name...

...

SP500.csv

...

If I show you its contents...

```
1 # read in the data
2 sp500 <- read.csv(here::here("presentation", "SP500.csv"))
3 head(sp500)
```

	observation_date	SP500
1	2021-01-20	3851.85
2	2021-01-21	3853.07
3	2021-01-22	3841.47
4	2021-01-25	3855.36
5	2021-01-26	3849.62
6	2021-01-27	3750.77

If I tell you where I got it from...



But I cannot give it to you!

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So how can you verify my results?

- By obtaining the file again
- By running my code again
- By verifying that the results are the same!

Trust but verify!

- Reproducibility is key!
- But there are weaknesses...

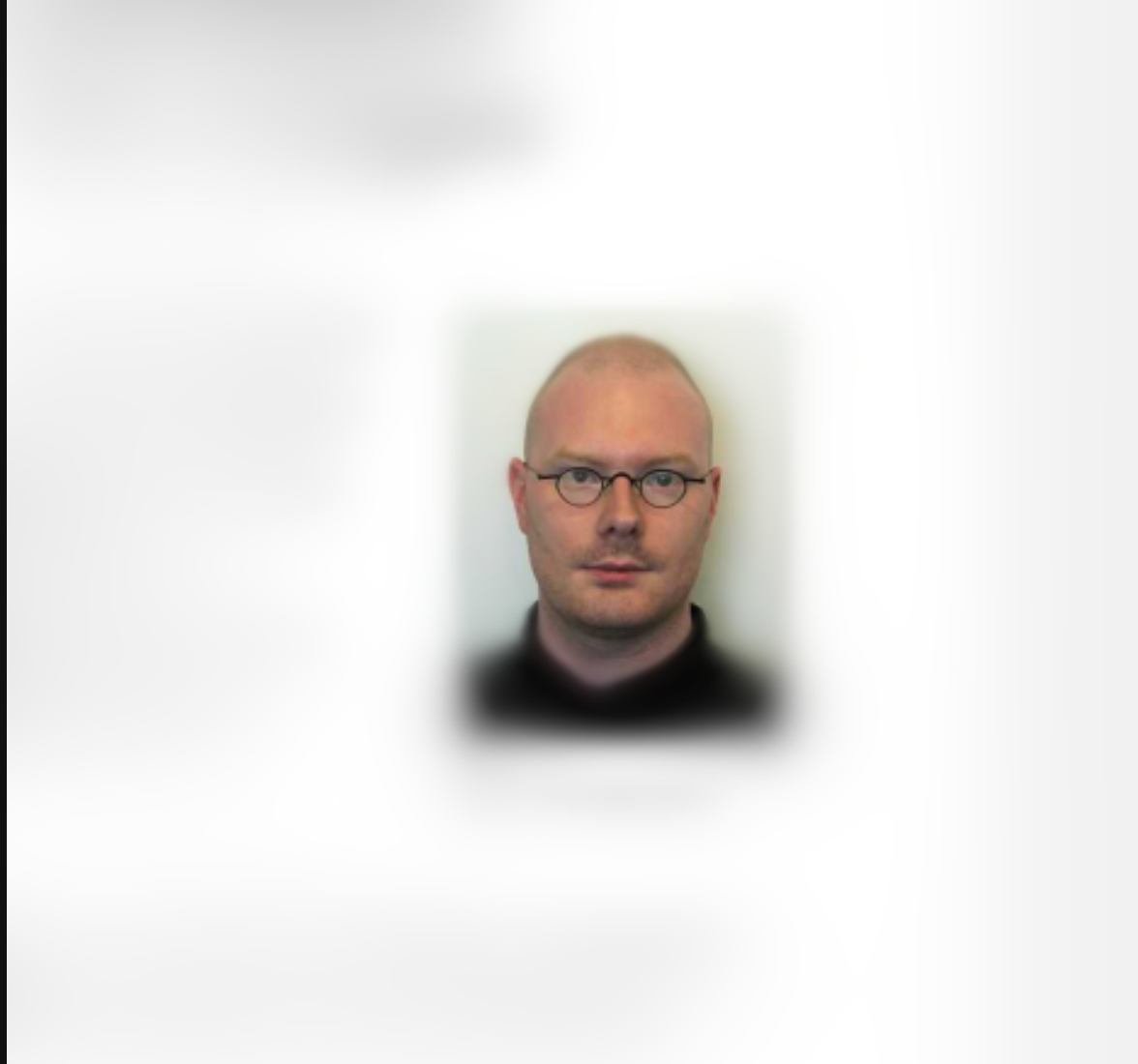
The issue of provenance

SP500

- Created by S&P Dow Jones Indices LLC
- Widely used as a market benchmark
- Commercial product, cannot be redistributed
- S&P considered **reliable**
- **Me** getting the file is the ***weak link***

**What if I am the
collector of data?**

Who is this person?



Dirk Smeesters

Following investigation, Erasmus social psychology professor retracts two studies, resigns

The social psychology community, already rocked last year by the Diederik Stapel scandal, now has another set of allegations to dissect. Dirk Smeesters, a professor of consumer behavior and society at the Rotterdam School of Management, part of Erasmus University, has resigned amid serious questions about his work.



Dirk Smeesters

According to an Erasmus press release, a scientific integrity committee found that the results in two of Smeesters' papers were statistically highly unlikely. Smeesters could not produce the raw data behind the findings, and told the committee that he cherry-picked the data to produce a statistically significant result. Those two papers are being re-

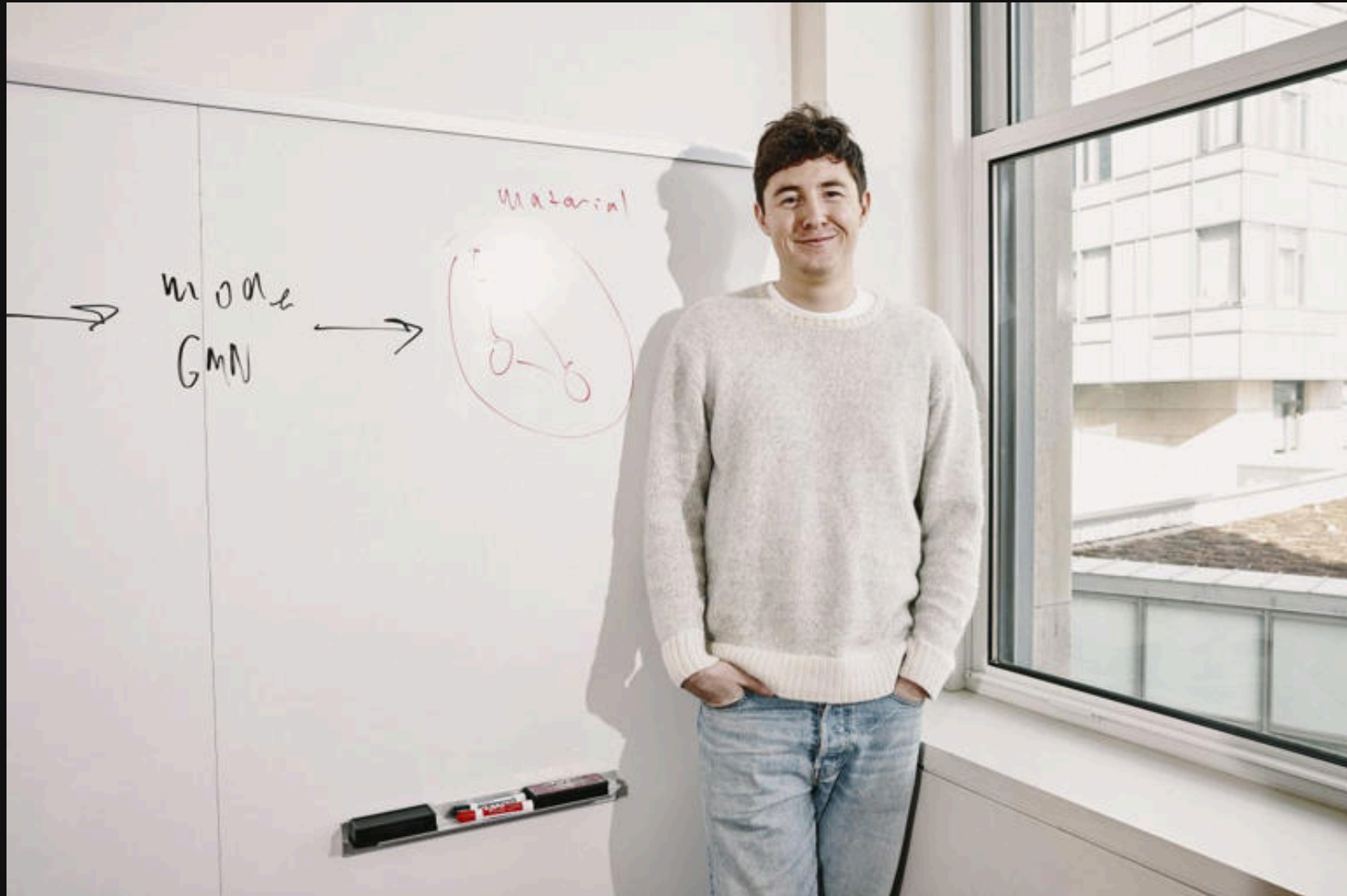
Retraction

“a scientific integrity committee found that the results in two of Smeesters’ papers were **statistically highly unlikely**. Smeesters ***could not produce the raw data*** behind the findings, and told the committee that he **cherry-picked** the data to produce a statistically significant result. Those two papers are being retracted, and the university accepted Smeesters’ ***resignation*** on June 21.”

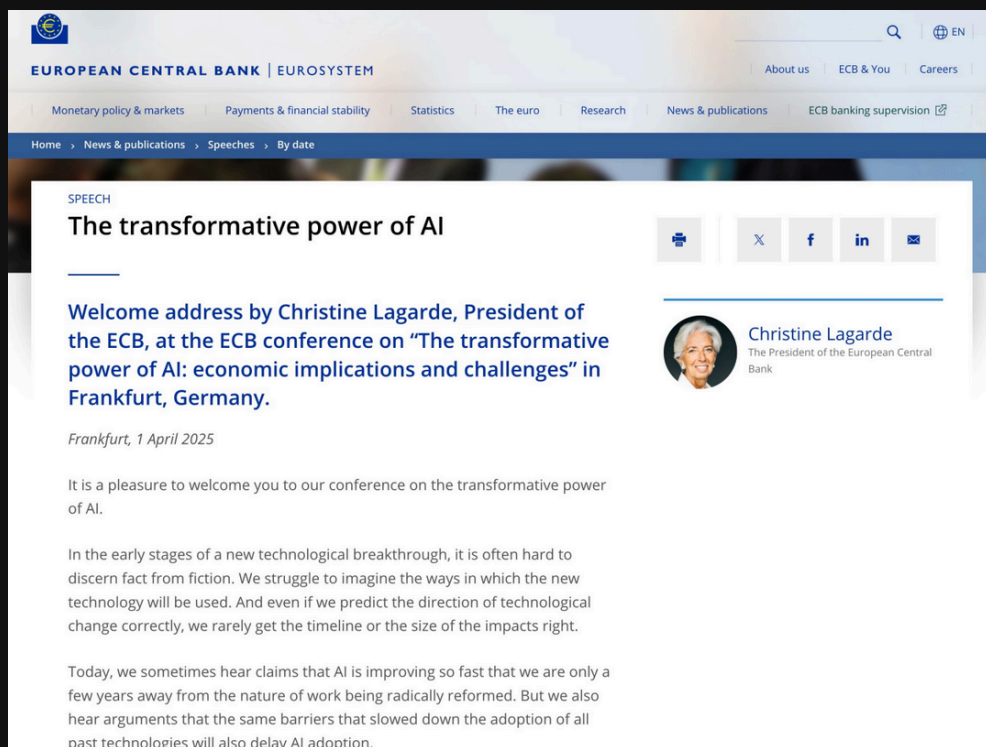
Who is this person? (2)



Aidan Toner-Rodgers



Maybe you've heard about him



"AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings."

ECB speech

Now

“MIT now declares “**no confidence in the provenance**, reliability or validity of the data and...in the veracity of the research”. Mr Toner-Rodgers’s paper has been **withdrawn** from the pre-print repository on which it first appeared [arXiv]; ... The lab at the heart of his findings remains **unknown**.”

**How can we know
that a data source is
reliably obtained?**

Consider the case of Gino

The New York Times

Account ▾

The Harvard Professor and the Bloggers

When Francesca Gino, a rising academic star, was accused of falsifying data — about how to stop dishonesty — it didn't just torch her career. It inflamed a crisis in behavioral science.



Francesca Gino



The case of Gino

- Francesca Gino was a **tenured** professor at Harvard Business School, writing on **honesty** (!)

The case of Gino



- Several articles were investigated by third parties (*Data Colada*, in particular ¹), and found to be problematic

The case of Gino

- At least one of them had manipulated data **AFTER** it had been collected, **BEFORE** it had been analyzed.

How The Moral Impurity Data Were Altered
In the Promotion Condition

Participant IDs in the Posted Data

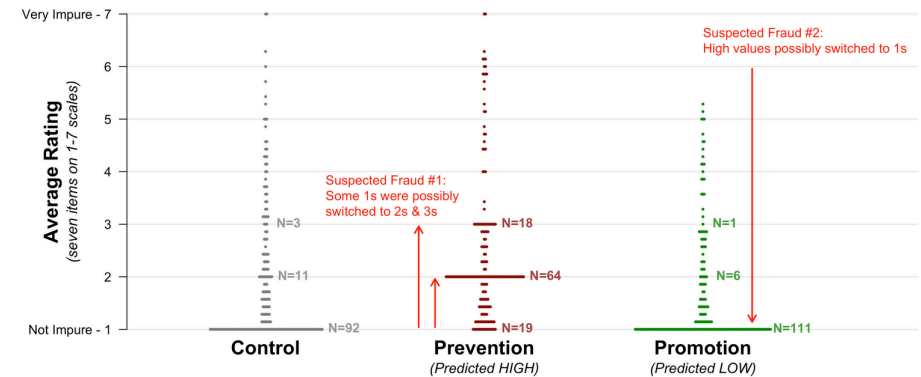
ID	Impure1	Impure2	Impure3	Impure4	Impure5	Impure6	Impure7
233	0	0	0	0	0	0	0
200	-2	-2	-2	-2	-2	-2	-3
447	-3	-3	-5	-3	-5	-6	-5
471	-3	-4	-5	-4	-5	-6	-5
335	-4	-3	-6	-4	-5	-6	-6
319	0	0	0	0	0	0	0
199	-2	-4	-3	-4	-5	-2	-4
30	0	0	0	0	0	0	0
498	-4	-3	-5	-2	-2	-4	-4
237	0	0	0	0	0	0	0
118	-5	-5	-6	-4	-4	-5	-5
120	-4	-4	-4	-4	-4	-4	-4
204	0	0	0	0	0	0	0
309	0	0	0	0	0	0	0
589	-4	-4	-5	-4	-5	-6	-5
220	0	0	0	0	0	0	0
251	0	0	0	0	0	0	0
248	-5	-5	-6	-4	-5	-6	-4
364	-6	-6	-6	-6	-6	-6	-6
376	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
268	0	0	0	0	0	0	0
290	-3	-2	-3	-2	-2	-4	0
47	0	0	0	0	0	0	0
454	-3	-2	-4	-3	-4	-4	-4
441	-4	-3	-2	-3	-4	-5	-3
538	-6	-6	-6	-6	-6	-6	-6
8	0	0	0	0	0	0	0

Participant 200's 7th moral impurity rating was decreased by 3

All of Participant 538's moral impurity ratings were decreased by 6

Average 'Moral Impurity' Rating For Each Participant

Gino, Kouchaki, & Casciaro (2020) -- Study 3a (Posted Data)



Results of manipulation

Data manipulation

What can **YOU** do?



What is this?



Training

- Biology students learn key lab techniques
 - Pipetting
 - Capture-recapture of wild animals

Training

- Biology students learn key lab techniques
 - Pipetting
 - Capture-recapture of wild animals



That's my daughter's hands in there

YOU!

I WANT YOU
FOR REPROCUCIIBLITY



(BTW...)

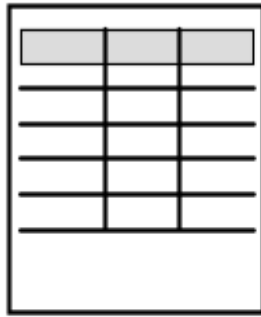
When I prompted Gemini to correct the spelling in “Reproducibility”, it only made it worse!

That looks fine, but correctly spell
reproducibility



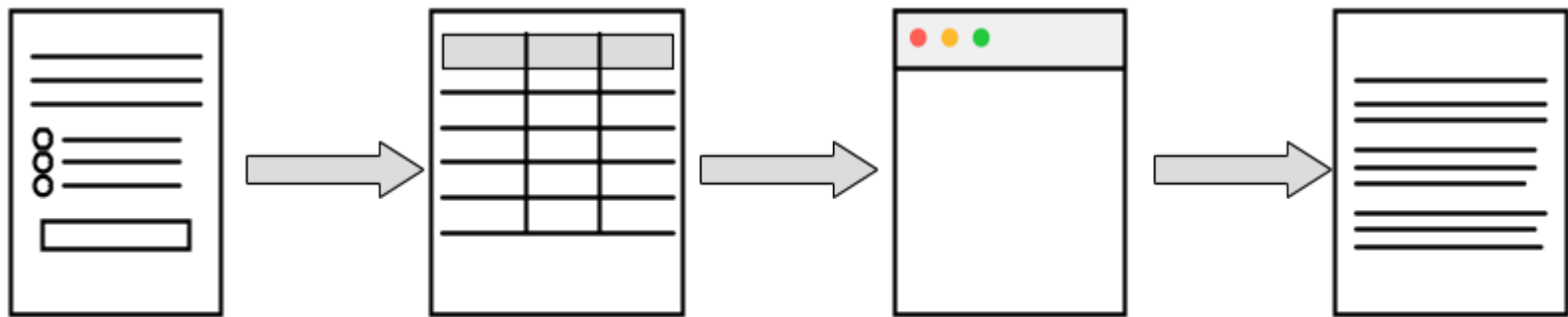
Back to the topic

Generic survey processing



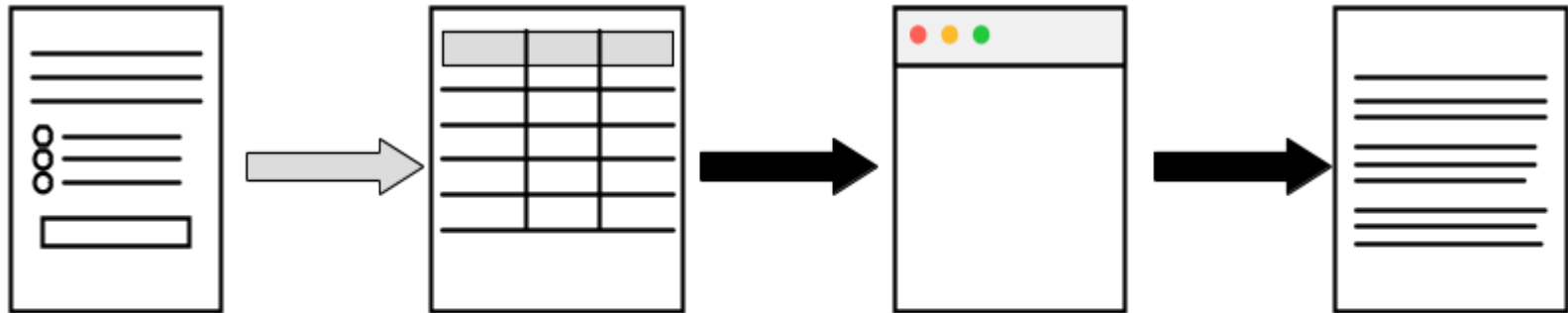
Generic survey processing

Generic survey processing



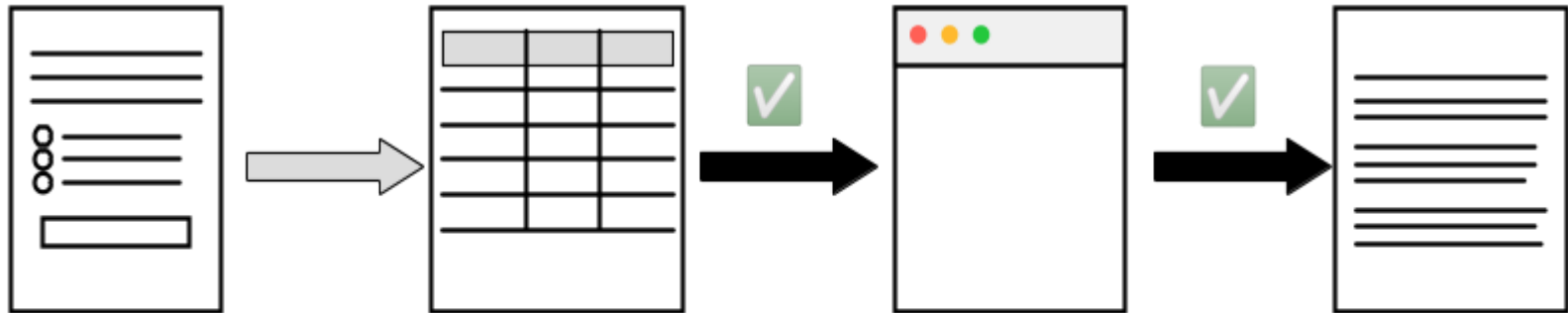
Generic survey processing

Requiring transparency in academia



Generic survey processing

Verifying transparency in academia



Generic survey processing

Verification by journals

- **Provision** (publication of materials) provides transparency
- **Verification** (running the analysis again - computational reproducibility) compensates for *mistrust/absence of trust*

Which journals

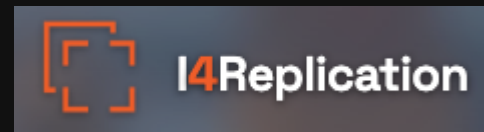
- American Economic Association (8)
- Econometric Society (3)
- Canadian Journal of Economics (1)
- Royal Economic Society (2)
- Western Economic Association International (1)
- European Economic Association (1)
- Review of Economic Studies (1)
- Journal of the European Economic Association (1)
- Journal of Political Economy (3)
- American Journal of Political Science (1)
- American Political Science Review (1)

Verification by others

- Pre-publication: **cascad**
- Post-publication: **Data Colada, Institute for Replication**



cascad



I4R

Verification by institutions

- World Bank²



Reproducible Research Repository

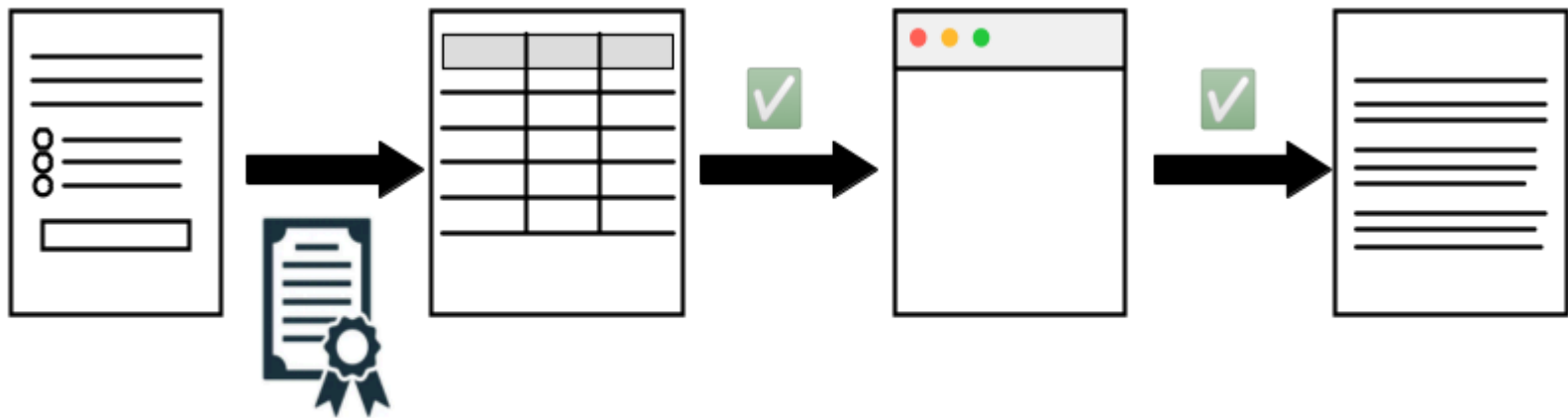
World Bank RRR

Verification by you



- Toner-Rodgers was caught b/c others scrutinized his work
 - Investigated the statistics (not too fishy)
 - Investigated the sources (huh?)
 - Investigated the coherence of the paper (much harder)

Taking it a step further



Survey flow

Taking it a step further

- Has been discussed by authors behind Data Colada
- Survey tool provider (Qualtrics, etc.) exports data, posts **checksum**
- Survey tool provider exports data only to institution directly into trusted repository, researchers obtain data from there (with privacy protections)
- Researcher can verify **checksum**

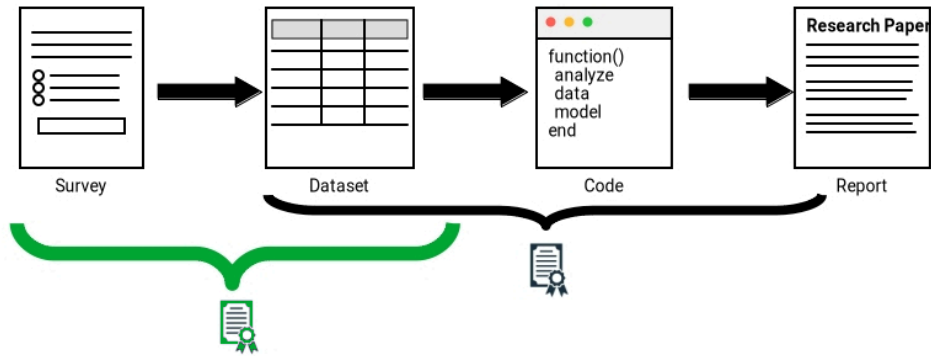
Does not work yet

What if you could verify the file?

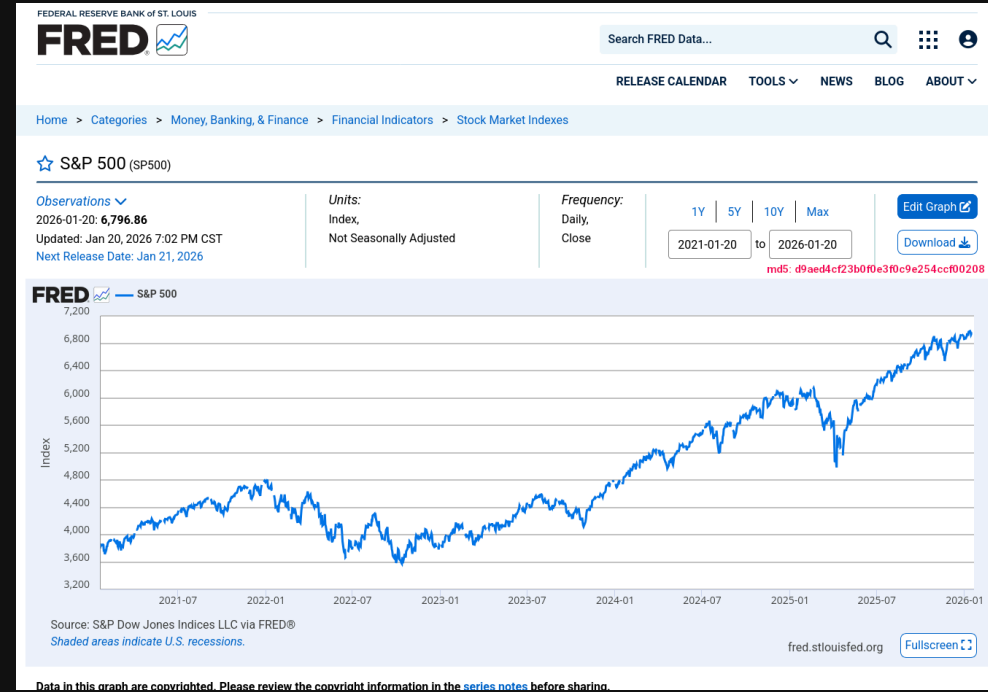
```
1 # compute checksum of the file  
2 tools::md5sum(here::here("presentation", "SP500.csv"))
```

```
/home/runner/work/presentation-2026-01/presentation-2026-  
01/presentation/SP500.csv
```

```
"d9aed4cf23b0f0e3f0c9e254ccf00208"
```



Certified survey processing



Not yet available!

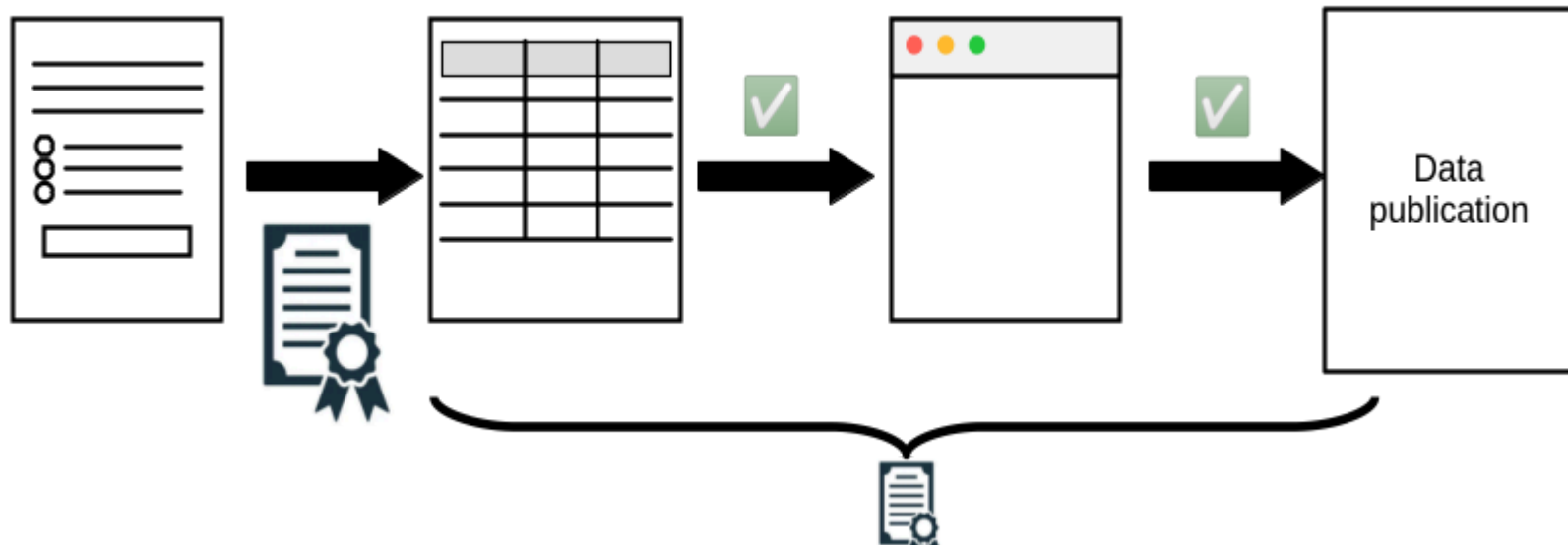
Does not prevent all fraud

Toronto researcher loses
Ph.D.

**Exclusive: Psychology
researcher loses PhD after
allegedly using husband in
study and making up data**

Toronto case

How to document the full process?



Survey flow

How to document the full process?

- Identify all sources - very precisely!
- Document all processing steps - using tools you are learning here!
- Transparent about your process - show your code!
- Expect critique (if not criticism) - embrace it!

Churchill said...



"Do not trust any statistics you did not fake yourself."

Or did he...



“No, of course, the British Prime Minister ... **never claimed such nonsense.** But putting his name in front of a quote gives it a more solemn, more imposing, more definitive appearance.” [Source]

A sketch: Transparency Certified

<https://transparency-certified.github.io/>

TRACE: Building trust in computational research

A new approach to computational transparency and reproducibility

Transparency Certified



Work in progress

- Working with **cascad**, several **INEXDA** members, and others
- Relying on external certification of data inputs (data catalogs with metadata, checksums)

Who is this person? (3)



Who is this person? (3)



Alp Simsek, one of ...

Transparency is the norm



nearly 5,000 authors who
have had their work
verified by the AEA Data
Editor and team.

You can do it, too!



**I WANT YOU
FOR REPROCUCIIBLITY**



You can do it, too!



Bonus: You can help, too!

- Apply the reproducible tools *not just for research*: Policy analysis, program evaluation, business analytics, etc.
- Push data providers to provide better access tools (**API access**, checksums, **DOI**)
- Be able to explain why this *benefits* them!

The end!



Source

- This document's source:
<https://github.com/larsvilhuber/presentation-2026-01>
- Licensed under  CC BY-NC 4.0

Footnotes

1.

<https://datacolada.org/109>, <https://datacolada.org/110>,
<https://datacolada.org/111>, <https://datacolada.org/112>,
<https://datacolada.org/114>, <https://datacolada.org/118>

2.

Jones, M. (2024). Introducing Reproducible Research Standards at the World Bank. *Harvard Data Science Review*, 6(4).
<https://doi.org/10.1162/99608f92.21328ce3>